

Topic 17C: Carboxylic acids
9. be able to identify the carboxylic acid functional group
10. understand that hydrogen bonding affects the physical properties of carboxylic acids, in relation to their boiling temperatures and solubility
11. understand that carboxylic acids can be prepared by the oxidation of alcohols or aldehydes, and the hydrolysis of nitriles
12. understand the reactions of carboxylic acids with: - i lithium tetrahydridoaluminate (lithium aluminium hydride) in dry ether - ii bases to produce salts - iii phosphorus(V) chloride (phosphorus pentachloride) - iv alcohols in the presence of an acid catalyst
13. be able to identify the acyl chloride and ester functional groups
14. understand the reactions of acyl chlorides with: - i water - ii alcohols - iii concentrated ammonia - iv amines
15. understand the hydrolysis reactions of esters, in acidic and alkaline solution
16. understand how polyesters are formed by condensation polymerisation reactions